

Assignment-3

Day 1: Tools for Troubleshooting Theory Overview of tools (multimeter, POST card, software diagnostics)

Task-1: List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue and explain in one line how the multimeter helps in each case.

	Real-life scenario	How the multimeter helps
1	PC won't turn on	Measures the SMPS output voltages to check whether the power supply is providing the correct voltage.
2	Laptop charger is not charging	Measures the DC voltage at the charger output to determine whether the adapter is supplying power.
3	Smartphone won't charge	Checks voltage and continuity around the charging port/cable to identify a broken connection or power-delivery problem.

Task-2: Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.

The POST card displays hexadecimal codes that show which stage of the motherboard's Power-On Self-Test is running.

In the demonstration, the card helps identify where the computer is getting stuck during startup.

The displayed code can be checked against the motherboard/POST-code table to identify the likely faulty component.

This makes it useful for diagnosing problems such as CPU, RAM, or graphics-related boot failures.

Task-3: Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.

Sensor	Value	Min	Max
CHAUDHARI_SAHAJ			
Intel Corporation 440BX Desktop Reference Platform			
Utilization			
Physical Memory Load	38.0 %	38.0 %	38.0 %
Available Physical Mem...	4.9 GB	4.9 GB	4.9 GB
Virtual Memory Load	26.1 %	26.1 %	26.3 %
Available Virtual Memory	7.3 GB	7.3 GB	7.3 GB
Intel Core i5 12450H			
Powers			
Power Max (PL1)	0.00 W	0.00 W	0.00 W
Short Power Max (PL2)	0.00 W	0.00 W	0.00 W
Clocks			
Cores (Max)	2493.0 MHz	2493.0 MHz	2493.0 MHz
CPU BCLK	2863311.5 MHz	2863311.5 MHz	2863311.5 MHz
Memory	2863311.5 MHz	2863311.5 MHz	2863311.5 MHz
Utilization			
Processor	8.7 %	7.2 %	17.0 %
Intel Core i5 12450H			
Powers			
Power Max (PL1)	0.00 W	0.00 W	0.00 W
Short Power Max (PL2)	0.00 W	0.00 W	0.00 W
Clocks			
Cores (Max)	2494.0 MHz	2494.0 MHz	2494.0 MHz
CPU BCLK	2863311.5 MHz	2863311.5 MHz	2863311.5 MHz
Memory	2863311.5 MHz	2863311.5 MHz	2863311.5 MHz
VMware Virtual NVMe Disk (VMware NVME_0000)			
Temperatures			
Assembly	30.0 °C	30.0 °C	30.0 °C
Counters			
Power-On-Hours	60 hrs	60 hrs	60 hrs
Power Cycle Count	10	10	10
Health Status	100 %	100 %	100 %
Utilization			
Used Space (c:)	38.0 %		
Available Capacity (c:)	39.0 GB		
Activity	0.0 %	0.0 %	3.6 %
Speed			
Read Rate	0.00 MB/s	0.00 MB/s	0.08 MB/s
Write Rate	0.09 MB/s	0.00 MB/s	3.48 MB/s
Read IOPS	0.00 KIOPS	0.00 KIOPS	0.00 KIOPS

Task-4: Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Tool	What it checks	Example Problem Detected
Multimeter	Checks electrical values such as voltage, resistance, and continuity .	Faulty SMPS/charger output or broken connection.
POST Card	Checks motherboard startup/POST codes during boot.	RAM, CPU, or motherboard failure preventing the PC from booting.
Software Diagnostics	Checks hardware status, temperatures, storage health, and system errors through software.	Overheating CPU or a failing HDD/SSD .